OpenSourceRail

Closing the full design capability gap

Repository review and engineering development roadmap
Mechanical | assembly | electrical | HVAC | civil | digital twin

Review baseline: commit 97773368, 19 October 2026, 19:21 London. The review covers the newest changes and a targeted assessment of the wider engineering architecture, analysis register, design generators, integration models and retained evidence. It is not a line-by-line audit of every repository file or a complete software release test.

Main conclusion

The project has developed a substantial engineering workbench: parametric geometry, native CAD correspondence, bounded numerical studies, manufacturing and erection scenarios, revision-bound evidence and automatic design proposals. The next step is to join these capabilities into complete, configuration-specific product definitions with shared load paths, interfaces, operating duty and verification contracts.

The newest assembly programme is a useful vertical slice, not a full railway designer. It covers two physical pilot families and two implementations of one offline cassette controller. The recommended strategy is to finish three reusable pilot chains, then extend verified component families across the product breakdown.

Start with	Why it matters	Required end-to-end evidence
Vehicle service cassette	Tests clip-in assembly across mechanical, wiring and ventilation	Detailed parts and interfaces; harness and duct design; insertion/removal; bench and installed tests
Vehicle structural bay	Connects frame, panels, equipment and real loads	Manufacturing-ready joints; fatigue/buckling/modal checks; mass/CG/inertia; repair and fire evidence
One viaduct bay	Links lightness to foundations, logistics and erection	Complete reactions and joint design; ground applicability; staged erection; transport; QA and installed cost

What was verified during this review

119 focused tests passed; three native-CAD tests were skipped locally because FreeCAD was unavailable. Both public evidence packages verified; the continuous-delivery videos decoded successfully. Published logs separately record 149 delivery tests and 35 assembly tests passing in their recorded environments. Those retained native checks were not rerun locally. Repository source was not modified.

1. Latest additions and actual coverage

The five latest commits repair delivery selection, exported-copy verification and publication recovery, correct Pi-specific requirement allocation, and introduce a sourced Baghdad assembly baseline with a bounded development programme.

Area	Verified current position	What remains
Publication	Delivery-policy and complementary leaders; cross-checks to raw exports; recovery tests for commit/push phases	Continue failure-injection tests and expose publication state in the common UI
Assembly campaign	38 trials: 18 cassette recipes, 18 bearing-seat recipes and two controller variants	The fixed domain exhausts; new families need generators and trusted evaluators
Cassette results	12 screened with open requirements; six failed	Locking, counterpart connector, termination, full electrical/HVAC duty and qualification remain open
Bearing-seat results	Nine failed; nine held; native geometry exists for a provisional representative	No bearing candidate closes all screens; exact offered bearing and full load path remain unresolved
Coverage	414 scoped records: 345 legacy/unreconciled plus 69 Baghdad records; 21 sourced facts	Three records have pilot generators; 411 lack a component-specific pilot allocation. These are not 411 proven distinct physical parts
Wider analysis register	30 entries: 18 screening and 12 planned	No calibrated, independently-checked or accepted entry is recorded in that register

Do not confuse maturity counts with component completion

The delivery ledger reports 340 representative unit revisions: 308 have defined/native-verified geometry and 14 have the scoped numerical screening flag. These map to five authoritative item records. Manufacturing-qualified units and accepted components remain zero. Repeated units and architecture alternatives must not be counted as distinct completed component designs.

Immediate load-transfer correction

assembly_design/evaluate.py computes bearing normal demand from deck mass and a uniformly apportioned train mass. It omits the known 60,000 kg track-support allocation in the reference kit. Equal division across eight bearings alone contributes about 73.6 kN per bearing before load factors; actual reactions require the complete structural model. Replace the parallel calculation with a revision-bound reaction envelope from the complete civil kit. Include rails, trackform, services, walkway, braking, thermal, uplift and erection actions with explicit combinations.

New baseline values include six-car planning cases of 384/460.8 tonnes and 18.5 m nominal car length. They must remain separate from LM3 and other vehicle definitions until an explicit architecture and interface baseline is adopted.

2. One controlled product and evidence model

Deliverables

• Freeze a named prototype configuration: vehicle geometry, low-floor and door layout, consist/bogie arrangement, passenger/duty cases, route/speed/gauge, charging concept, climate and emergency operating states. Record sensitivities separately from adopted requirements.

• Reconcile the 345 legacy and 69 Baghdad source records by architecture and physical function. Preserve source IDs; identify duplicates, alternatives, assemblies and requirements. Group repeated parts into reusable generator/evaluator families instead of writing one adapter per record.

• Use one typed component contract: owned versus purchased item; revision; material/process; solids; mass, CG and inertia; datums; interface endpoints and degrees of freedom; tolerances; loads and units; manufacturing operations; inspection characteristics; uncertainty; provenance and applicability.

• Add interface contracts for structural mounts, bearings, electrical connectors, fluid/air ports and software signals. Each change declares which CAD, analyses, harnesses, schedules, costs and tests it invalidates.

Consolidation approach

Keep discipline solvers as adapters. Share identities, recipes, receipt verification, qualification and publication across design_engine, continuous_design, assembly_design, twin_workflow and generative_design. Migrate incrementally behind compatibility interfaces; do not launch another independent workflow or attempt a wholesale rewrite.

The existing assembly contract and design_consistency representations are a suitable starting point. Use FreeCAD for editable detailed geometry and drawings; use IFC/Bonsai for civil placement, classification, property requirements and coordination. Exchange models must preserve the same IDs; neither exchange format becomes a second design authority.

Source and standards contracts

Every source card should carry publisher, exact part/grade, revision, URL, document hash, locator, units, licence, retrieval date, operating envelope and uncertainty. Separate maximum ratings from derated performance; processing temperature from continuous service; monthly weather averages from design extremes; reference material properties from batch-qualified allowables.

Maintain a licensed standards/edition register with jurisdictional adoption and applicable vehicle/equipment categories. IEC 61373 now has a 2026 edition; perform an impact review rather than silently inheriting older references. IEC 62928 addresses lithium-ion traction batteries, so another chemistry needs an explicit applicability and test strategy. [R18-R23]

Exit evidence

A change to one mount, connector or material regenerates every affected representation and requests the relevant analyses/tests. A deliberately mismatched LM3/Baghdad configuration, stale supplier document or missing interface must be rejected. IFC IDS checks enforce required information, while engineering solvers check performance. [R01, R17]

3. Mechanical design and material systems

Close the structural and functional load paths

• Develop the underframe, side/roof structures, floor transitions, doors/windows, equipment supports and bogie/articulation interfaces as connected assemblies. Replace gross-section shortcuts with local models at holes, welds, inserts, cut-outs and load introduction points.

• Generate reversible joints with bolt stack, engagement, preload/locking basis, bearing/net-section/block-shear/prying checks, corrosion and access. Add weld throat, detail category, sequence, distortion allowance and NDT plan. CAD thread detail is optional where a controlled specification and load model are sufficient; connection behaviour cannot be optional.

• Analyse global bending/torsion, local buckling, fatigue spectra, modal transfer, lifting, jacking, recovery, attachment loss and degraded configurations. Geometry-free crash energy allocations are requirements inputs, not crashworthiness verification; the crash load path needs its own justified model and tests.

• For bogies, wheelsets, brakes, traction and articulation, select qualified purchased subsystem boundaries initially. Require interfaces, mass/CG/inertia, stiffness/damping, braking/traction maps, wheel profiles, fatigue envelopes and maintenance constraints. Localise frames, brackets and tooling only when manufacturing and testing capability is demonstrated.

GFRP and lightweight materials

Extend the existing laminate/shell tools to local reinforcement, inserts, edge closures, adhesive thickness, ply drops, core shear/crushing, delamination, impact damage, hot/wet ageing and repair. Model the complete resin/fibre/core/bond/finish system and process. A higher-temperature processing core or UL94 rating alone does not qualify continuous hot-roof operation or rail fire performance. Preserve the current failed 85 C roof cases until compatible data and tests exist. [R06, R07, R23]

Tools and output package

Reuse FreeCAD/build123d, Gmsh and CalculiX; add verified shell/contact/preload and nonlinear adapters where necessary. Use topology optimisation only inside a process-constrained domain, then reconstruct manufacturable CAD and independently remesh it. NASA fastener guidance and CalculiX examples are implementation hints, not adopted rail acceptance criteria. [R01-R05]

Every pilot should export editable CAD, neutral STEP, issued-state drawing data, material/process specification, BOM, joint register, tolerances, assembly/removal instructions, inspection plan and exact analysis inputs. Bought-out internals may remain supplier-controlled when their configuration and verified performance are sufficient.

Exit evidence

Complete one structural bay including panel and equipment attachments. Reconcile CAD/BOM/measured mass, demonstrate mesh and boundary-condition sensitivity, compare an independent formulation, and correlate representative joint and panel tests. No mass saving may come from omitted brackets, seals, cables or fasteners.

4. Assembly, clip-in modules and DfMA

Complete the service cassette first

The new cassette already represents tray, rail walls, mount pads, hardware, duct walls, conductors, jackets and a connector envelope. Finish the mating connector, cable terminations, clamp lining and retention, earth conductor, duct seals, inspection cover and installation interface before expanding its topology search.

• Use guided location followed by positive retention: slide-in runners, datum seats, captive fasteners, locking pins or a qualified latch. Define indication of fully seated/locked state and secondary retention where required. Treat insertion force, release force, mis-mating and wear as design variables.

• Distinguish bench subassembly, transport restraint, temporary mounting, final lock, electrical/fluid connection, installed tests and maintenance removal. A component fitting in its final position does not prove that it can be installed.

• Replace generic operation lists with a shared joint/process graph: endpoints, dependencies, fixtures, tool and human access, insertion/rotation path, fastening/curing parameters, test prerequisites and teardown. Generate schedule, QA and cost data from it.

• Check worst-case dimensional stacks and thermal movement first; add statistical tolerance/yield analysis only when process distributions are measured. Include weld distortion, composite cure variation, site survey error and connector float in the interface stack.

Apply the same method to civil kits

Use standardised piles/cages, foundation caps, columns, cap/bearing seats, deck units, track panels and walkway/service cassettes. Civil structural connections require qualified force transfer, temporary stability and durability. Self-locating features simplify erection but do not replace engineered anchorage, prestress or grout design. [R14]

Measure assembly outcomes

Compare conventional installation, guided captive-fastener assembly and a suitable quick-lock option. Record factory/site person-hours, critical path, tool changes, working-at-height exposure, misassembly, first-pass yield, rework, repair time, lifts and complete installed cost. Benchmark at the intended production volume; extra connectors, rails and locks can increase mass and purchase cost.

Exit evidence

A bench-tested module is installed and removed within a demonstrated access envelope. Its serial/revision links to test results; lock, connection, bonding and seal checks occur after installation. Missing fastening, an impossible removal path, a disconnected service or a changed interface invalidates the relevant result. Qualify the module with vibration/thermal/environmental conditions applicable to its mounting location. [R18]

5. Electrical, batteries and energy interfaces

Move from allocations and pin lists to electrical design

• Create hierarchical single-line and schematic definitions for traction, charging, battery/BMS, converters, auxiliaries, LV distribution, protection, earthing/bonding and communications. Define nets, nominal and limiting voltages, cable lengths, fault return paths and isolation boundaries.

• Generate harness drawings and manufacturing BOMs from the same netlist: both connector halves, terminals, seals, backshells, strain relief, crimps, wire type/gauge, shielding, routing, bend radii and identification. WireViz is a harness-documentation tool, not a full electrical system verifier. [R09]

• Calculate continuous/transient current, voltage drop, cable/connector thermal derating, inrush, precharge, fault current, protective-device coordination, insulation/clearance, bonding and EMC architecture. Purchased power connectors need applicable simultaneous current/temperature data; headline maxima are insufficient.

Exploit the exact supplier detail already found

The referenced Harwin connector page supplies a technical drawing, mating/pin information, component and cable-assembly specifications, assembly/removal instructions and test reports. It uses loose solder contacts and jackscrews; the complete kit, mating part and termination process should be represented. Generic crimp assumptions must not be applied to this solder-contact part. CAD download access may require supplier registration. [R24]

Battery, charging and traction duty

Bind the existing battery/PyBaMM toolchain to the selected cell chemistry and measured parameter set. Calculate repeated station-to-station cycles including actual acceleration, gradient, passengers, HVAC, braking recovery, charging time, charger concurrency, losses, hot/aged cells and degraded modes. Verify power, energy and temperature independently; a one-minute charging aspiration is a constraint to test, not a guaranteed capability. [R10, R11, R19]

Represent coolant flow, loss-of-cooling, leak/vent routing, contactors/fuses/sensors, HV interlock, emergency isolation and fire interfaces. Selected motor, inverter and brake maps must establish tractive effort, stopping duty and thermal limits. Traction battery requirements do not automatically cover sodium-ion or other substitute chemistries.

Tools and exit evidence

Reuse existing energy, pandapower/pvlib and battery studies; add KiCad/ngspice adapters for circuit-level transients and electrothermal models. Validate one complete LV cassette and one battery/charger/auxiliary chain with independent power/heat balances, intentional open/short/reversed/mis-mated faults, supply variation, and applicable SIL/HIL tests. [R08-R11]

6. HVAC, cooling and air quality

Define the complete duty before selecting modules

Create one whole-car model for sensible and latent loads: passengers, solar absorption, glazing, infiltration/door cycles, equipment, ventilation, climate histories and hot-soak. Separate ambient air, sunlit roof temperature and equipment temperatures. Add humidity, fresh air/CO2 strategy, filter loading, condensate, noise, pressure and degraded-mode requirements. The new duct cassette is ventilation packaging, not a refrigeration design.

Model in increasing fidelity

• Level 1: upgrade the existing one/two-zone heat/moisture and duct-loss models with controlled duty and component maps. Enforce mass, energy and water conservation.

• Level 2: couple fan curves, duct network, dampers, filters and refrigerant/coolant circuit models. Use Modelica/Buildings or Modelica Fluid with railway boundary conditions, and CoolProp for humid-air/refrigerant properties. These tools are reusable modelling components, not rail-specific qualification. [R12, R13]

• Level 3: apply OpenFOAM to shortlisted duct layouts and passenger-zone distribution, including bypass, stagnation, obstruction and thermal boundaries. Require mesh/time convergence and comparison with flow/temperature measurements. CFD should confirm selected designs, not consume the budget for every coarse candidate. [R15]

Physical interfaces and manufacturing

Complete flange/coupling geometry, seal compression, insulation/vapour barrier, drain gradients, access doors, duct supports, compressor/fan isolation, fire/smoke penetrations and service replacement. Include their mass and auxiliary demand. Mechanical locking and electrical interlocks must use the same cassette states and serial/revision identity.

Failure and climate cases

Test full occupancy, hot start, frequent doors, restricted filter/duct, failed fan or cooling module, sensor bias and loss of cooling supply. Recalculate battery/auxiliary capacity for every HVAC variation. Model supplier performance at relevant ambient and supply conditions, rather than constant cooling capacity or COP.

Exit evidence

A complete pilot car or representative cabin/duct rig demonstrates airflow, distribution, temperature/humidity response, condensate control and auxiliary demand. Use the applicable EN 14750 framework through a controlled standards register. Rail Tec Arsenal and rail HVAC suppliers offer reference test practices and potential later test facilities; no current availability or quotation is implied. [R20, R31]

7. Civil structures, foundations and erection

Complete the reference bay as a physical kit

• Generate actual manufactured reinforcement cages, prestress/tendon/anchor details, diaphragms, bearings, jacking/replacement seats, lifting inserts, drains, track fasteners, parapets and services. Keep unresolved allowances visible until they are replaced by owned detail or verified purchased items.

• Use one signed load/reaction envelope from the selected train and civil configuration for deck, bearings, caps, columns and foundations. Correct the bearing-seat omission before optimisation. Do not recompute a separate generic force for each component.

• Add nonlinear cracking/ultimate/shear/torsion/fatigue, creep/shrinkage, prestress transfer and losses, bearing restraint/rotation, thermal movement, uplift, seismic/wind and staged construction as applicable. Detailed local effects and resistance are separate from global elastic equilibrium.

Ground and urban constructability

Replace uncalibrated equivalent springs with site-zoned strata, groundwater, investigation coverage, settlement and pile-group checks; implement p-y/t-z/q-z and cyclic response where justified. Candidate pile and shallow-foundation families must satisfy actual resistance, serviceability, utilities and construction-vibration constraints. OpenSees pile examples and JRC worked examples are implementation references, not Baghdad soil data. [R16, R25]

Optimise the complete system

In the retained 20 m/two-support reference, piles, cages, caps and piers account for roughly 73% of total kit mass. Deck-only lightening therefore has limited leverage. Compare hollow/tapered columns and caps, compatible pile routes, span length, pier spacing and joint families together with material choices. Retain resistance and durability constraints when exploring lightweight concrete, high-performance sections or hybrid details.

Extend isolated-bay comparisons to repeated corridors with shared supports, route-specific crossings/stations, plant movement, mould reuse, transport permits, crane capacity at working radius, outrigger reactions, temporary works, weather and working calendars. Compare whole-span lifting and segmental erection without assuming the lighter lift is the cheaper system.

Exit evidence

One kit has current-source verified load paths and independent checks for permanent and construction states; transport and erection can be replayed; connection mock-ups and material/ground inputs are traceable; as-built survey and installation tests close the relevant interfaces. FHWA prefabricated-connection guidance is valuable for tolerance, grout and temporary support planning. [R14]

8. Full vehicle and train-track-structure integration

Use the substantial models already present

integration_programme and civil_exploration already contain spatial/flexible vehicles, joints, wheel/rail contact and full-passage campaigns. Some retained full-train evidence is explicitly historical and source-mismatched. Reuse and requalify those adapters instead of inventing a second physics framework or treating all existing dynamics as absent.

• Bind generated body/frame modal mass and modes, supplier suspension/joint laws, bogie/wheelset properties and real wheel/rail profiles to the exact mechanical configuration. Include changed equipment mass, CG and inertia, not just gross mass added to axle loads.

• Connect civil-kit-study, bearing-seat and vehicle cassette revisions to a controlled integration interface. The current automatic integration still selects civil-detail rather than the new complete kit.

• Cover full approach, passage and exit; critical speeds, curves, cant, roughness, traction/braking, crosswind and degraded suspensions as applicable. Track wheel unloading/contact-domain limits, deck/rail reactions, clearance, carbody/seat accelerations and bearing/foundation demand.

• Run independent time-step, spatial mesh, rail discretisation and modal convergence. A short diagnostic kernel, solver completion or timed-out refinement is not a completed passage verification.

Couple energy and control as well as mechanics

HVAC and auxiliaries affect battery temperature and traction reserve; voltage and communication faults affect equipment permission; braking regeneration affects electrical limits. Use explicit power/heat/force interfaces with agreed causality and sample times. Prefer an adapter contract and repeatable co-simulation harness; add another framework only for a demonstrated gap.

Calibration and independent evidence

Create separate parameter, calibration and holdout datasets. Use analytical controls and independent numerical formulations first, then measured joint/modal/bogie/track tests. A second AI critic or the same solver at a finer mesh provides useful scrutiny but does not independently establish material laws or physical validity. [R26]

Exit evidence

A changed physical part triggers a fresh full-system analysis using its actual mass/modal/interface effects. Reaction and energy ledgers reconcile; required passages and adverse cases complete; independent checking and applicable physical correlation are linked to the same revision.

9. Generative design and bounded AI development

Separate two recursive loops

Design loop: requirements and interfaces -> generate candidate -> regenerate geometry/BOM -> inexpensive screens -> multidisciplinary evaluation -> retain Pareto alternatives -> independent/native confirmation -> evidence-based research adoption or rejection.

Capability loop: uncovered requirement -> scoped generator/evaluator work order -> isolated developer proposal -> separate critic -> independent benchmark/adverse controls -> reproducible registration -> bounded design search. A reviewed proposal must not silently become a trusted evaluator.

Move beyond the current 38 recipes

• Register families for joints, brackets, panels, structural bays, harness/duct modules, battery enclosures, bearing seats, pile/cap/pier units and station modules. Every family declares its valid geometry/material/process and load domain, outputs and independent checks.

• Expand shapes through a controlled grammar: stock/folded/welded members, rib patterns, shell thickness, laminate schedules, core grades, local reinforcements and assembly partitions. Represent their manufacturing constraints before exploration.

• Keep material properties in inseparable sourced bundles. Process temperature, fibre direction, density, strength, fatigue, fire and hot/wet properties cannot be independently picked to manufacture an artificial optimum.

• Retain objectives for complete installed mass, cost, factory/site effort, largest lift, delivery time, replacement effort and uncertainty. Required inspection and performance coverage remain constraints. Unknown costs remain null; operation counts are not calibrated labour.

Efficient and reproducible search

Use fast analytical models for broad exploration and detailed FEA/CFD for shortlisted candidates. Record validity domains, uncertainty and discrepancy between fidelities. Preserve failed/adverse candidates, seeds, budgets and exact tool/source identities. Cross-validate surrogate models against untouched cases and verify the final manufactured CAD, rather than an unreconstructed topology mesh. [R05, R26]

Allocate effort to under-covered families and important constraints; stop repeated trials when the obstacle is missing data or an unimplemented model. Publish the actual policy leader and complementary alternatives consistently across every dashboard and export.

Exit evidence

A benchmarked family automatically generates, tests and improves both a shape choice and an assembly/material choice; known invalid candidates are rejected; an intentionally misleading objective or relaxed requirement cannot win. Improvements are attributed to closed requirements and demonstrated margins, not to trial count.

10. Testing, qualification and physical correlation

Tier	What must be checked	Minimum evidence
Contract/property	Units, identity, architecture, missing data, revision impact, source bounds	Schema/property/mutation tests; missing or nonfinite values cannot pass
Analytical/reference	Beam, bolt group, circuit, thermal/flow and soil controls	Independent worked cases, conservation, adverse/boundary controls
Native geometry	Solids, joints/interfaces, manufacturing IDs and neutral exchange	FreeCAD reopen/STEP; volume/mass/CG/inertia; clash and path checks distinguished
Numerical detail	Local stress/contact, shells, buckling, electrothermal, CFD, nonlinear ground	Mesh/time/boundary sensitivity; independent formulation; domain checks
Assembly/system	Fault propagation, installation, full duty and complete train passage	Exact revision integration; mass/power/heat/reaction ledgers; missing-step rejection
Physical correlation	Coupons, joints, subassemblies, car/bridge and target hardware	Calibrated instruments, raw data, fixture identity, uncertainty, holdout prediction
Production/installed	Factory acceptance, transport damage, site interfaces and commissioning	Serial/batch traveller, traceable records, disposition/retest and authority release

Strengthen CI without repeating low-value checks

Keep fast contract/adverse tests on ordinary changes. Run native CAD/FEA on affected generators and scheduled complete acceptance profiles on known equipped runners. A skipped required native check should leave the relevant design gate open. Pin the tested solver/runtime profile; distinguish archived evidence verification from a fresh physics run.

Add the bearing dead-load omission as an adverse load-ledger test. Add tests for new clip/connector interfaces, electrical open/short/incorrect mate, duct leak/blockage, local panel damage, prestress sequence, ground-domain violations and final assembly removal. Maintain publication recovery and copy-consistency regressions.

Acceptance data interface

Implement authenticated ingestion of raw test records, calibration certificates, equipment/fixture revision, method/standard edition, operating conditions and measurement uncertainty. Require explicit applicable review and disposition. Existing serial/revision QA event handling is a good base; hashes and reviewer-name strings do not authenticate an external measurement.

Use a test pyramid: inexpensive material and joint coupons first; then representative cassette/panel/bearing assemblies; then car, bogie and civil demonstrators. Choose tests to reduce the largest uncertainty and verify safety-critical interfaces before building a full prototype. [R26-R32]

11. Delivery work packages and dependencies

WP / lead	Deliverable	Dependencies / exit
00 Systems	Prototype architecture, duty, environment and standards/adoption baseline	Start now; resolve contradictory dimensions/masses and map alternatives
01 CAD/data	Common component, joint, source, evidence and family contracts; legacy crosswalk	00; propagate one change through CAD, analysis, QA and costs
02 Loads/integration	Shared mass/CG/inertia, force/reaction, power/heat and interface ledgers	00-01; correct bearing loads and reject missing allocations
03 Mechanical	Complete structural bay, panels, equipment interfaces and bought-out boundaries	01-02; global/local checks, drawings and testable joints
04 Manufacturing	Completed cassette and civil kit process graphs; insertion/removal and QA sequence	01-02; real trial assembly/removal and calibrated effort
05 Electrical	Traction/LV/charge schematics, harnesses, protection, battery duty and cooling	00-02; circuit/energy/thermal faults plus bench/HIL evidence
06 HVAC	Whole-car duty, flow/refrigeration maps, duct/seal/drain/control design	00-02,05; convergence and thermal/flow test correlation
07 Civil/ground	Complete bay and support units, connections, ground and erection states	00-02; load path, construction checks and mock-up evidence
08 Dynamics	Exact flexible train/track/kit co-simulation with full passages	02-07; converged domain-valid cases and independent checks
09 Generative	Family grammar, multi-fidelity search and benchmarked capability registration	01 plus each discipline; final exact CAD and non-mass Pareto leaders
10 V&V/QA	Layered benchmark/CI system and authenticated test/correlation ingestion	Starts with 01; governs every discipline exit
11 UI/ERP/cost	Unified coverage, comparison and release views; manufacturing/QA handoffs	01,04,10; exact BOM/revision/serial and complete scope pricing
12 Prototype team	Manufacture/test the three pilot chains; resolve NCRs and calibrate models	Applicable design exits; repeatable build and independent accepted evidence

Software work packages are not proof of physical completion. Assign a named owner and reviewer to each WP and a unique evidence bundle to each exit. Group component records into families before estimating effort. Do not promise closure of hundreds of records by multiplying a two-assembly pilot.

12. Proposed sequencing and practical first actions

First 2 weeks: establish a coherent development baseline

Complete WP00 and the minimal WP01 contracts. Fix the bearing reaction interface. Reconcile the three runner/report views and architecture names. Create family coverage and source/standards registers. Freeze acceptance cases for the cassette, structural bay and viaduct bay. Obtain exact supplier drawings/maps and identify missing site/test inputs.

Weeks 3-6: make the pilots complete enough to build and test

Deliver the mating/locking and harness/duct details, structural bay joint package and civil bearing/pier-cap interface. Add circuit and thermal-fluid adapters with independent controls. Generate actual assembly/removal paths, fixtures and inspection characteristics. Build calibrated bench fixtures and coupon programmes. Publish uncertainty and open inputs alongside each result.

Weeks 7-12: integrate and calibrate

Run complete duty and train-to-kit cases, correlate selected physical tests, and measure production/assembly effort. Demonstrate one controlled generative improvement per pilot with final native confirmation. Finish prototype RFQ/build packs and cost/production comparisons. Longer material, fatigue, fire, geotechnical and railway qualification campaigns continue beyond this software/pilot window.

Team and ownership

Provide systems/configuration leadership; mechanical/composites; electrical/battery; thermal-fluid; civil/geotechnical; vehicle dynamics; CAD/BIM automation; manufacturing/QA; and V&V/software capabilities. Roles may be combined or part-time, but a developer/AI critic pair does not replace independent domain review or appointed engineering authority.

Repository organisation

Keep canonical component definitions in the existing component catalogue and discipline models in engineering. Put small recipes, source records and benchmark controls in Git; retain large raw histories under bounded evidence storage with portable review exports. Consolidate common queue/evidence/publication code incrementally. Maintain one index pointing to current source, latest current-source result and historical evidence.

Useful common UI

Show architecture, requirements coverage, owned/purchased scope, missing geometry/interfaces, source applicability, evidence maturity and next blocking action. Compare mass/lift/assembly/time/cost with assumptions visible. Clicking a part should reveal its joints, drawings, analysis cases and QA traveller. Distinguish planned tests, executed checks, physical correlation and accepted records. Preserve Arabic/RTL and offline field usability already addressed elsewhere in the project.

13. Completion contracts for the three pilot chains

Pilot	Digital specification complete when	Physical/production evidence still required
Service cassette	Every owned part, mating interface, cable and duct connection is defined; wire/connector derating and airflow duty are checked; tools and removal paths work; BOM and instructions agree	Measured pull/lock/earth/leak/flow tests; applicable vibration/environment/fire results; controlled factory and installed acceptance
Structural bay	Load cases, materials, joints, panel inserts and equipment mounts are complete; current local/global/modal/fatigue checks and drawings are linked; assembly and repair are defined	Material/process coupons; joint/panel/subassembly correlation; weld/bond and production capability; applicable crash/fire and vehicle validation
Viaduct bay	Actual permanent and erection load paths reach ground; all forces/reactions include installed items; bearings/anchors/reinforcement/lifting details, logistics and QA are complete	Site investigation/survey; qualified materials and connection mock-ups; lift/temporary-work acceptance; surveyed as-built and installed tests

A proposed automatic decision gate

A design revision may enter a research shortlist only when every implemented mandatory check passes and unknown applicability is visible. A digital-specification gate additionally requires all required parts/interfaces/operations and analyses to be represented, with no silent allowances for critical connections. Physical and manufacturing qualification then require their own applicable authenticated records. These states should be independently queryable.

What additional websites can and cannot close

Public manuals, worked examples, source code and supplier documents can close many implementation gaps: formulas, API adapters, detail patterns, parameter domains and test methods. They cannot supply actual Baghdad soil/survey, an offered product configuration, batch properties, local workshop capability, quotations or witnessed performance. Store those as external input work packages, not as modelling failures or an excuse to lower requirements.

For low cost, prioritise local repeatable cutting/forming/welding, reusable moulds, simple fixtures and replaceable modules. Purchase specialist bearings, connectors, braking and battery subsystems where qualification economics favour them. Country of manufacture is a procurement field, not evidence of fitness or a substitute for lifecycle cost.

PRIMARY SOURCES CHECKED 19 OCTOBER 2026

Technical references 1

R01 | FreeCAD Assembly Workbench
https://github.com/FreeCAD/FreeCAD-documentation/blob/main/wiki/Assembly_Workbench.md
Use native assembly joints and degrees of freedom as geometry/interface hints. Add OSR load, retention and verification contracts; an assembly constraint is not a strength calculation.

R02 | CalculiX
https://calculix.de/
Structural FE documentation and examples. Use validated contact, shell and bolt-preload models; independently check each adapter and mesh/domain assumptions.

R03 | Gmsh reference manual
https://gmsh.info/doc/texinfo/
Automated meshing and physical-group export. Preserve component/interface IDs and support repeatable local refinement and mesh-quality gates.

R04 | NASA RP-1228 Fastener Design Manual
https://ntrs.nasa.gov/citations/19900009424
Fastener selection and joint-design background. Use worked cases as independent benchmark hints; this aerospace manual does not adopt railway criteria.

R05 | CalculiX BESO
https://github.com/calculix/beso
Open topology-optimisation implementation. Add process constraints, minimum features, reconstruction, remeshing and exact manufactured-CAD verification before selection.

R06 | Diab datasheets and certificates
https://www.diabgroup.com/certificates-datasheets/
Candidate core property bundles and processing data. Request continuous-service, hot/wet, fatigue and fire applicability; do not equate processing temperature with service qualification.

R07 | Gurit industrial composite engineering
https://www.gurit.com/industrial/
Potential laminate, tooling and material-system guidance. Request current compatible resin/fibre/core/bond data and process/test scope; historic rail examples do not qualify OSR.

R08 | KiCad schematic and simulation manual
https://docs.kicad.org/9.0/en/eeschema/eeschema.html
Hierarchical electrical schematics and ngspice integration. Freeze the project-supported version; generate checks from OSR nets and applicable supplier models.

R09 | WireViz
https://github.com/wireviz/WireViz
Text-based harness drawings, pinouts and harness BOMs. Document individual harnesses; independently check ampacity, protection, mating compatibility and full-system connectivity.

Links are research hints and possible information sources. Public methods, product-family claims and supplier maxima must not be copied into accepted OSR requirements without applicability checks. Some normative texts and CAD downloads require licences or registration.

Technical references 2

R10 | ngspice electrothermal tutorial
https://ngspice.sourceforge.io/ngspice-electrothermal-tutorial.html
Examples of electrical-loss and thermal-network feedback. Benchmark converter/contact/heatsink models and preserve their validated operating domain.

R11 | PyBaMM parameterisation examples
https://docs.pybamm.org/en/pybamm-v26.10.0.0/source/examples/notebooks/parameterization/parameterization.html
Parameterisation methods and provenance. The repository pins another version: adapt and verify against that version before upgrading. Use chemistry-specific measured data.

R12 | LBNL Modelica Buildings library
https://simulationresearch.lbl.gov/modelica/
Open dynamic thermal-fluid/control models and validation examples. Adapt to railway duty, ventilation, door events and supplier performance; building assumptions are not rail criteria.

R13 | CoolProp
https://coolprop.org/
Fluid/refrigerant and humid-air properties for coupled HVAC/cooling models. Property routines do not select a compliant refrigerant or qualify an installation.

R14 | FHWA prefabricated bridge connection guidance
https://www.fhwa.dot.gov/bridge/prefab/if09010/01a.cfm
Connection tolerances, grout, handling and assembly planning. Highway practice is a detail/test-method reference; use adopted railway loads and project criteria.

R15 | OpenFOAM heat-transfer solver documentation
https://doc.openfoam.com/2306/tools/processing/solvers/rtm/heat-transfer/
Buoyant flow and conjugate heat-transfer implementation hints. Select and pin an OpenFOAM distribution; solver names and interfaces differ across releases.

R16 | OpenSeesPy laterally loaded pile example
https://openseespydoc.readthedocs.io/en/latest/src/pile.html
Accessible p-y/t-z/q-z modelling example and references. Calibrate layers, groundwater, group/cyclic effects and validity to the actual project.

R17 | IfcTester / IDS
https://docs.ifcopenshell.org/ifctester.html
Validate IFC information requirements and export reports. Enforce IDs, required material/interface/QA properties; this is not structural-performance verification.

R18 | IEC 61373:2026 catalogue
https://webstore.iec.ch/en/publication/68999
Current published edition found during review for rolling-stock equipment shock/vibration. Confirm project adoption, mounting category and licensed test details.

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PRIMARY SOURCES CHECKED 19 OCTOBER 2026

Technical references 3

R19 | IEC 62928:2017 catalogue
https://webstore.iec.ch/en/publication/31101
Onboard lithium-ion traction battery design/operating/test scope. Do not assume automatic coverage of other chemistries or auxiliary battery systems.

R20 | BS EN 14750:2024 project record
https://standardsdevelopment.bsigroup.com/projects/2022-00612
Urban/suburban/regional rolling-stock air-conditioning comfort and type-test framework. Obtain adopted normative text and define OSR duty and climate categories.

R21 | BS EN 12663 vehicle-body reference
https://knowledge.bsigroup.com/products/railway-applications-structural-requirements-of-railway-vehicle-bodies-locomotives-and-passenger-rolling-stock-and-alternative-method-for-freight-wagons-1
Candidate vehicle-body structural framework already referenced by the new baseline. Freeze edition, category and project adoption; metadata does not supply load combinations.

R22 | COSQC technical services and standards
https://www.cosqc.gov.iq/index.php/service/all
Iraqi standards, technical requirements, conformity and consultation entry point. Confirm project-specific jurisdiction and approval routes with the responsible authorities.

R23 | BS EN 45545-2 fire behaviour reference
https://knowledge.bsigroup.com/products/railway-applications-fire-protection-on-railway-vehicles-requirements-for-fire-behaviour-of-materials-and-components-2
Candidate rail-material fire framework. Obtain applicable normative requirements and assess the actual installed stack, location and classification.

R24 | Harwin exact connector and documentation
https://www.harwin.com/products/M80-4000000F1-10-326-00-000
Technical drawing, mating/pin information, C005XX/C049XX component/cable specifications, IS-52 assembly instructions and test-report links. Verify the exact offered mating and termination configuration.

R25 | JRC bridge design worked examples
https://publications.jrc.ec.europa.eu/repository/handle/JRC68415
Basis/actions, structural, fatigue, geotechnical and seismic worked examples. Use as independent-method benchmarks; editions and national/project adoption need review.

R26 | NIST digital twin credibility paper
https://www.nist.gov/publications/credibility-consideration-digital-twins-manufacturing
Verification, validation and uncertainty across a digital twin lifecycle. Use to structure model credibility and applicability, not just evidence hashing.

R27 | University of Huddersfield IRR facilities
https://research.hud.ac.uk/institutes-centres/irr/facilities/
UK rail dynamics, wheel-rail, bogie and HIL capability. Ask for a narrowly scoped independent review and rig/correlation plan before a full vehicle campaign.

Links are research hints and possible information sources. Public methods, product-family claims and supplier maxima must not be copied into accepted OSR requirements without applicability checks. Some normative texts and CAD downloads require licences or registration.

PRIMARY SOURCES CHECKED 19 OCTOBER 2026

Technical references 4

R28 | NCC verification services
https://www.nccuk.com/services/technical/engineering-services/verification/
UK composite test-pyramid, material/environmental, NDT and verification support. Seek a coupon/insert/panel programme compatible with intended local manufacture.

R29 | UK Battery Industrialisation Centre
https://www.ukbic.co.uk/
Coventry battery development, manufacturing/assembly and testing entry point. Confirm selected chemistry, pack and test applicability, scope and commercial access.

R30 | NPL dimensional metrology
https://www.npl.co.uk/products-services/dimensional
UK measurement traceability, calibration and dimensional metrology. Useful for datum/tolerance and factory inspection capability planning.

R31 | Rail Tec Arsenal services
https://www.rta.eu/en/service
Vienna climatic wind-tunnel and vehicle/component climate testing. Consider later whole-car/HVAC correlation after low-cost bench and subsystem work.

R32 | Iraqi Ministry of Planning / COSQC
https://mop.gov.iq/en/central-organization-for-standardization-and-quality-control
Official Iraqi description of standards/calibration/conformity functions. Identify applicable accredited local laboratories and legal requirements; railway authorisation remains a separate scope.

R33 | HUBNER articulation systems
https://www.hubner-group.com/en/products/articulations/articulation-systems-for-trams/
Potential subsystem interface and supplier-document source. Request exact yaw/pitch/roll, loads, stiffness/damping, fatigue and maintenance envelopes; do not infer a localisation agreement.

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Potential locations for missing evidence

Organisation / location	Useful first enquiry	Expected output to request
Huddersfield IRR, UK	Vehicle/track dynamics, wheel-rail, bogie/brake HIL and model correlation [R27]	Scoped review, parameter/measurement requirements, test rig plan and raw data rights
NCC, UK	Composite inserts, hot/wet coupons, manufacturing process, NDT and panel test pyramid [R28]	Applicable material/process allowables, fixture/test method and uncertainty; confirm facility location
UKBIC, Coventry, UK	Selected battery chemistry/module/pack and thermal/charging measurements [R29]	Cell maps, impedance/thermal/ageing data, assembly/testing scope and costing
NPL, UK	Datums, tolerances, measurement uncertainty and calibration [R30]	Inspection capability study and calibrated measurement plan
Rail Tec Arsenal, Vienna, Austria	Later rail HVAC/whole-car climate validation [R31]	Applicable duty and test configuration, instrument plan, measured response and report scope
COSQC / appointed local labs, Iraq	Applicable standards, calibration, materials and conformity route [R22,R32]	Named applicable authority/laboratory scope and authentic records
Qualified Iraqi precast and fabrication shops	Actual mould/cure, reinforcement, lifting, transport and assembly trials	Time studies, capability/tolerance/reject data, load charts, inventory and complete quotes

Request a data package, not a generic endorsement

Every enquiry should identify component/configuration/revision; exact tests and operating envelope; sample/fixture and calibration requirements; required raw data and uncertainty; test/analysis report; rights to publish or retain evidence; responsibilities and commercial scope. Supplier or laboratory capability pages establish neither availability nor OSR qualification.

Site-specific evidence needed in Iraq

Appoint competent teams for surveyed alignment and utilities, ground investigation and groundwater, seismic/wind/flood exposure, lifting/transport routes, construction access and field acceptance. Existing Iraqi manufacturing capacity is valuable, but actual qualified product slots, capabilities and costs must be established for the selected design.

Recommended first decision

Approve a pilot scope and its verification contract, allocate the named discipline owners, then complete the service cassette and the shared bearing reaction interface. Expand the automatic design engine only as each reusable family gains independent evidence. This is the shortest credible route from a broad workbench to complete, testable designs.